system, but that operation is in no way the same as acting on the state with the operator **L**. For example, if the state of the system before we do the measurement is $|A\rangle$,it is *not* correct to say that the measurement of **L** changes the state to $\mathbf{L}|A\rangle$.

To make sense of this, let's look closely at an example. Fortunately, the spin example of the previous subsection is just what we need. Recall Eqs. 3.12:

$$\sigma_z|u\rangle = |u\rangle$$

$$\sigma_z|d\rangle = -|d\rangle.$$

In these situations, there is no trap because $|u\rangle$ and $|d\rangle$ are eigenvectors of σ_z. If the system is prepared in, say, the $|d\rangle$ state, a measurement will definitely give the result −1, and the σ_z operator transforms the prepared state into the corresponding post-measurement state, $-|d\rangle$. The state $-|d\rangle$ is the same as $|d\rangle$ except for a multiplicative constant, so the two states are really the same. No problems here.

But now let's review the action of σ_z on the prepared state $|r\rangle$, which is *not* one of its eigenvectors. From Eq. 3.19, we know that

$$|r\rangle = \frac{1}{\sqrt{2}}|u\rangle + \frac{1}{\sqrt{2}}|d\rangle.$$

Acting on this state-vector with σ_z gives the result

$$\sigma_z|r\rangle = \frac{1}{\sqrt{2}}\sigma_z|u\rangle + \frac{1}{\sqrt{2}}\sigma_z|d\rangle$$

or